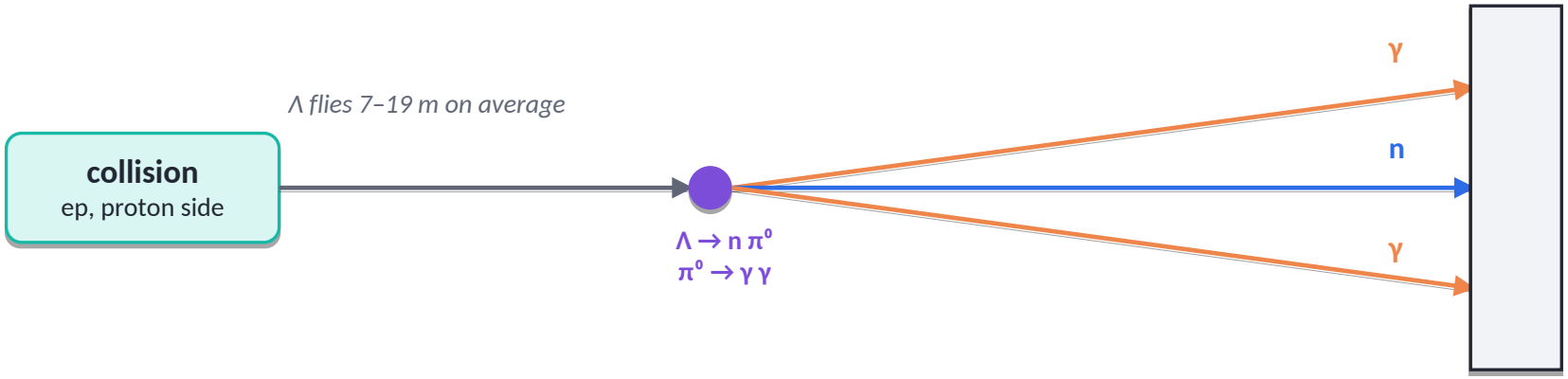
ZDC autoresearch, run 1: what went well and what went wrong

An agent-run experiment loop on the reconstruction of $\Lambda \to n\pi^0$ in the ePIC Zero Degree Calorimeter

Run of 2026-11-04/05, 22 hours · deck v1, built from the run's result files and ledger
For physicists outside the ZDC and outside agent-run loops: every term is defined where it is used

The task: $\Lambda \rightarrow n\pi^0$ at the Zero Degree Calorimeter, 36 m from the collision



- **ZDC: a block of absorber and scintillator tiles**
  - square tiles 4.9 cm, read out as 2.4 cm subcells
  - records where energy was deposited, layer by layer
  - accepts particles within 4 mrad of the beam (±14 cm)

35.8 m
collision point to the ZDC face

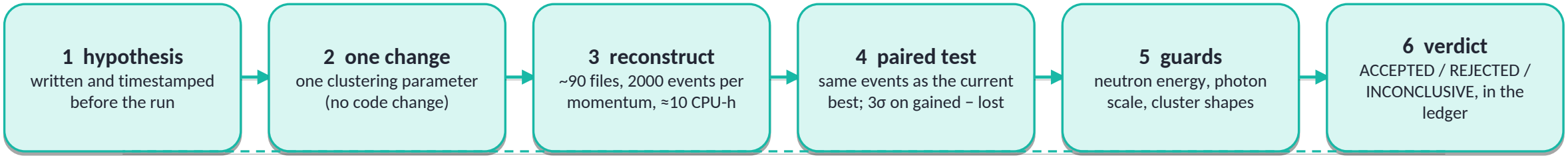
1.6–33 %
of $\Lambda \rightarrow n\pi^0$ reach the ZDC with all three particles (100–275 GeV)

4.5 % → ~0
Λ reconstruction efficiency after the 2026 change to square tiles

- **What the reconstruction has to do**
  - group hit subcells into clusters: one cluster per particle, then pair the two photons into a π^0 and the π^0 with the neutron into a Λ
  - the three showers overlap on the ZDC face; the clusterer joins neighbouring subcells, so one chain of hits becomes one cluster
  - the 2026 tile change widened the neighbour distance: the three showers now chain into one cluster and the Λ count fell to ~0
  - a proposed fix (EICrecon PR 3064) shrinks the distance; the loop's job: keep n, γ, γ in separate clusters without breaking the neutron measurement

Three showers on one calorimeter face after 36 m of flight: telling them apart comes before any Λ can be counted.

Autoresearch here: an agent executes a written loop, a human answers at checkpoints



Checkpoints

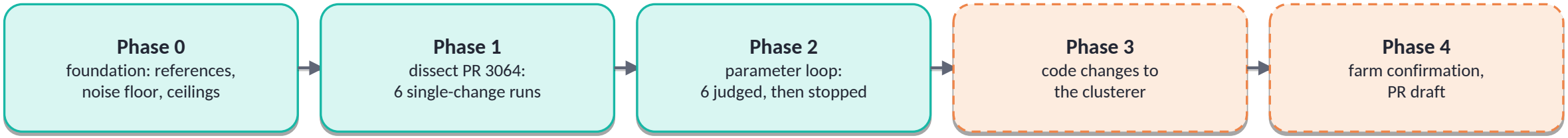
- every 5 experiments: a held-out sample (fresh seeds, never used to choose), a report, questions to the user
- after 10: stop for review

Frozen rulers

- the scoring scripts and the metric limits cannot change inside a phase
- a change to a ruler is a decision, logged with its reason

The record

- ledger: hypothesis, numbers, verdict for every run, rejected ones included
- decision log with the user's words; questions file; findings on code outside the program



The human sets the rules and the limits; the agent runs the loop and writes the record.

Run 1 in numbers: 22 hours, 15 reconstructions, 12 judged experiments

22 h
wall clock, 01:26 → 23:40 UTC; user offline for the first 12

141 CPU-h
reconstruction: 1333 files in 15 reconstruction rows

12
judged experiments: 2 accepted (one configuration), 8 rejected, 2 inconclusive

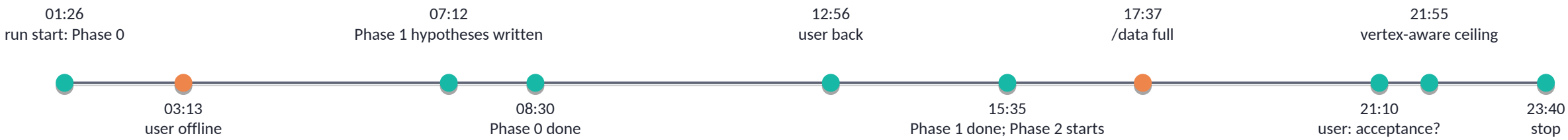
9
defects found in code the program does not own

21 + 25
decisions logged: user + agent defaults

8
questions put to the user, all answered

865 GB
peak disk use; 63 GB per experiment

2
parameter families closed by their guards (chaining, survival)



Timeline, UTC. Phases 0–1 ran unattended; Phase 2 with the user online.

Metrics, part 1: what the clustering does to each particle (S0–S3)

metric	measured on	the question it answers	role
S0γ, S0n, S0all	$\Lambda \to n\pi^0$ events, 8 momenta	is each photon, the neutron, all three, in a cluster of its own? (separated: the cluster is ≥ 80 % that particle and holds ≥ 50 % of its clustered energy)	selector: S0all decides
S1a	lone $\pi^0 \to \gamma\gamma$, 7 momenta	are the two photons of a π^0 in two clusters? — the simplest separation, no neutron in the way	diagnostic
S1b, S1c, S1d	lone π^0	do the two clusters give back the π^0 mass and energy? how precisely is a photon's impact point measured?	diagnostic
S2	lone photon, 8 energies	is the photon's energy right (scale), how wide is the spread (resolution), is it recognised as a photon (γ-tag)?	guard on the scale: within 5 % after a refit
S3	lone neutron, 6 energies	how much of the neutron's energy ends up in clusters (raw sum), in how many clusters, and is it ever mistaken for a photon?	guard: raw sum drop ≤ 5 %, clusters per neutron ≤ +0.5, mistag

- **Reading the rows**
  - every number is a fraction of events (or of particles) at one momentum; 1000–2000 events per point, so ±0.01 is resolvable
  - 'truth-matched' means the simulation's record of which particle made each hit is used to judge a cluster; a reconstruction on real data has no such record
  - 'raw' energy is the summed cluster energy before any calibration, so a calibration change cannot hide a lost shower

S0 asks the one question the clusterer controls; S2 and S3 ask whether the single particles are still measured right.

Metrics, part 2: the Λ outcome, physics-event guards and bookkeeping (S4, S5, G, T)

metric	measured on	the question it answers	role
S4a	$\Lambda \to n\pi^0$ events	in what fraction of events does the code output a Λ built from the right three clusters, with the right mass? (beside it: any Λ at all)	outcome: reported, never decides
S4b, S4c	$\Lambda \to n\pi^0$ events	how sharp is the Λ mass peak? how many photon-like objects per event (fragments make fake pairs)?	guards
S5	simulated physics events (MSF Λ sample)	Λ candidates per 1000 real-looking events, and how many of the true Λ are found	the physics goal; confirmation only
G1, G2, G3	deep-inelastic events (the CI's samples)	do clusters still look like showers (not single columns or odd shapes)? are the high-energy clusters still there? how much low-energy debris per event?	guards: limits set by the user
G4	every dataset	did anything outside the ZDC change? (every other collection compared file by file)	gate: a difference rejects
T	every dataset	reconstruction seconds per 1000 events	report line; 2× slower is a question

- **Reading the rows**
  - G1–G3 exist because the reviewers of PR 3064 saw three defects on physics events: single-column clusters, a spike in a shape angle, missing high-energy clusters; the guards make those measurable
  - S5 is the number the physics programme cares about; with 23 true Λ in 5000 events it cannot rank settings (slide 10)

The Λ numbers are the goal; they are reported beside every experiment but never allowed to decide it.

Why these metrics: one decides, several protect, the rest report

One decides: S0all

- it measures exactly what the clusterer controls: whether showers end up in separate clusters
- it counts events, so the same events can be compared setting against setting (gained − lost), which is far more sensitive than comparing two averages
- it is truth-matched: 'two clusters' alone could be one photon split in two

Several protect: S2, S3, S4b–c, G1–G4

- each guard is something the current reconstruction does well that a photon-friendly setting could silently break: the neutron's energy, the photon scale, shower shapes on physics events, everything else in the event
- the limits were set by the user before the loop ran; a loop that sets its own limits can always pass

The rest report: S4a, S5, S1, T

- S4a and S5 are the goal, and cannot decide: they are 0 for most settings, and they depend on the pairing code downstream of the clustering
- S1 isolates the photon–photon case; T flags a slowdown
- units added mid-run: per thrown $\Lambda \rightarrow n\pi^0$, so the 1.6–33 % acceptance is folded in

- **What a student should take from this**
  - a selector must be sensitive, must measure the thing being changed, and must be hard to game; the guards cover what the selector ignores
  - a proxy selector is a bet: that improving it improves the goal; the bet has to be checked with a number tied to the goal (next slide)
  - the limits are as much a physics statement as the selector: 'the neutron may lose 5 %' is a decision the physicist owns

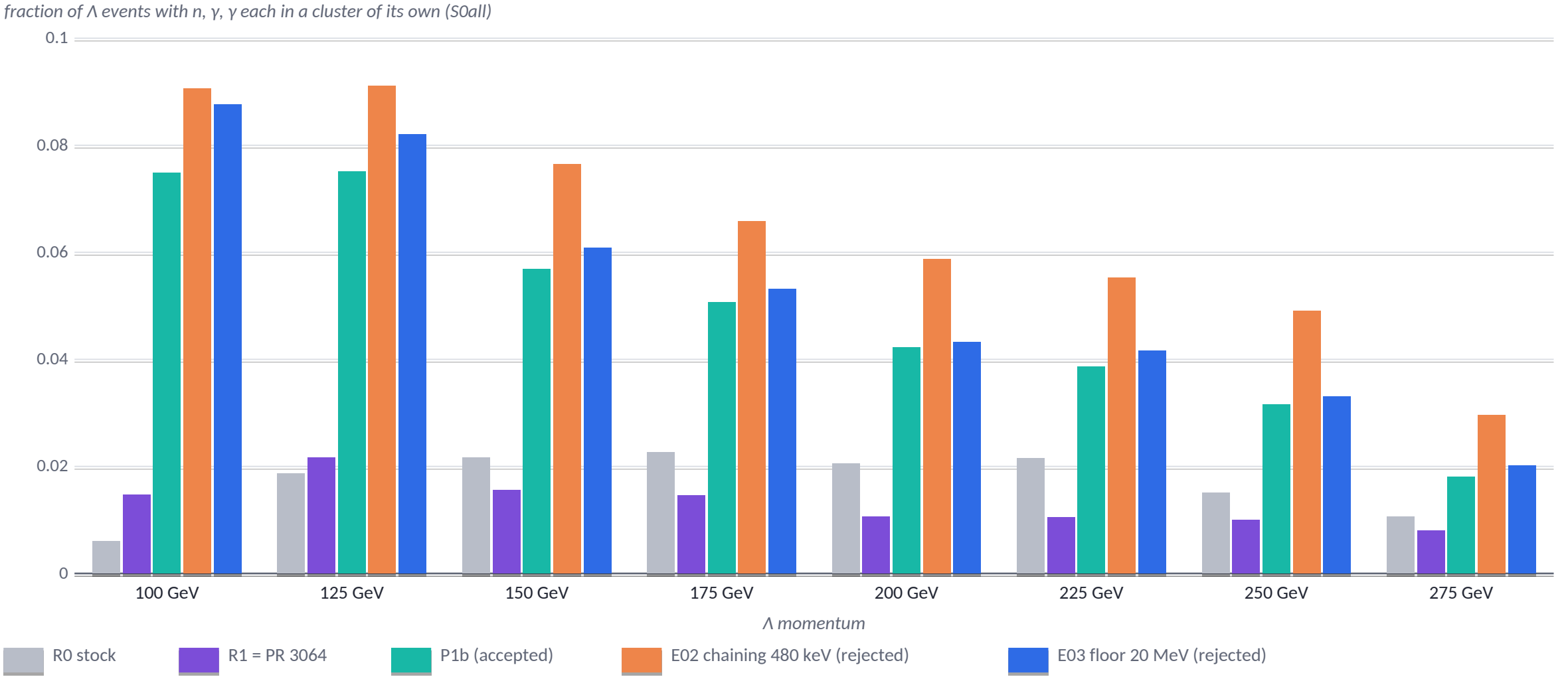
S0all is sensitive, on target and hard to game; the tie to the goal is what it lacks.

Metrics that were missing, for the loop and for reporting

missing	what it would have told us	status
a ceiling for separation (give each subcell to the particle that put most energy in it; score with the same rule)	how far each setting is from the best any clustering could do	never computed
a Λ number that moves while S4a is 0 (a vertex-aware pairing run offline on real clusters)	whether a separation gain changes the Λ yield at all	computed once, at the end, for perfect clusters only
who merges with whom: photon with photon, or photon with neutron	which code change would help: splitting showers or shortening the reach	the views say 'merged', not with what
separation by decay position, as a number	where the loss is geometry and where it is the clusterer	a plot per setting, no scoreboard cell
per-thrown-Λ units (acceptance folded in)	numbers comparable across momenta and with the physics analysis	added mid-run (user)
a prediction hit rate per checkpoint	whether the loop understands the detector or only screens	computed for this deck
cost per experiment: CPU-h, GB, wall time	the budget spent per answer; the disk would not have filled	T per 1000 events only
enough Λ for S5 and the mass peak (S4b)	23 true Λ in 5000 events cannot rank anything	farm-scale sample planned
a one-line report per experiment	a result readable without the ledger	added mid-run (user)

The first two rows are the gap that let the loop optimise a proxy for a day; both are afternoon-sized scripts.

S0all by Λ momentum: stock, PR 3064 and the three Phase 2 settings



Every 3×3 setting beats stock by 2–5× and stays below 0.1; PR 3064 is below stock above 125 GeV.

Per thrown Λ → nπ⁰: separation 0.3 → 0.6 %, Λ reconstructed 0 %, ceiling 14 %

All three particles separated, mean over 100–275 GeV



Λ reconstructed, mean over 100–275 GeV

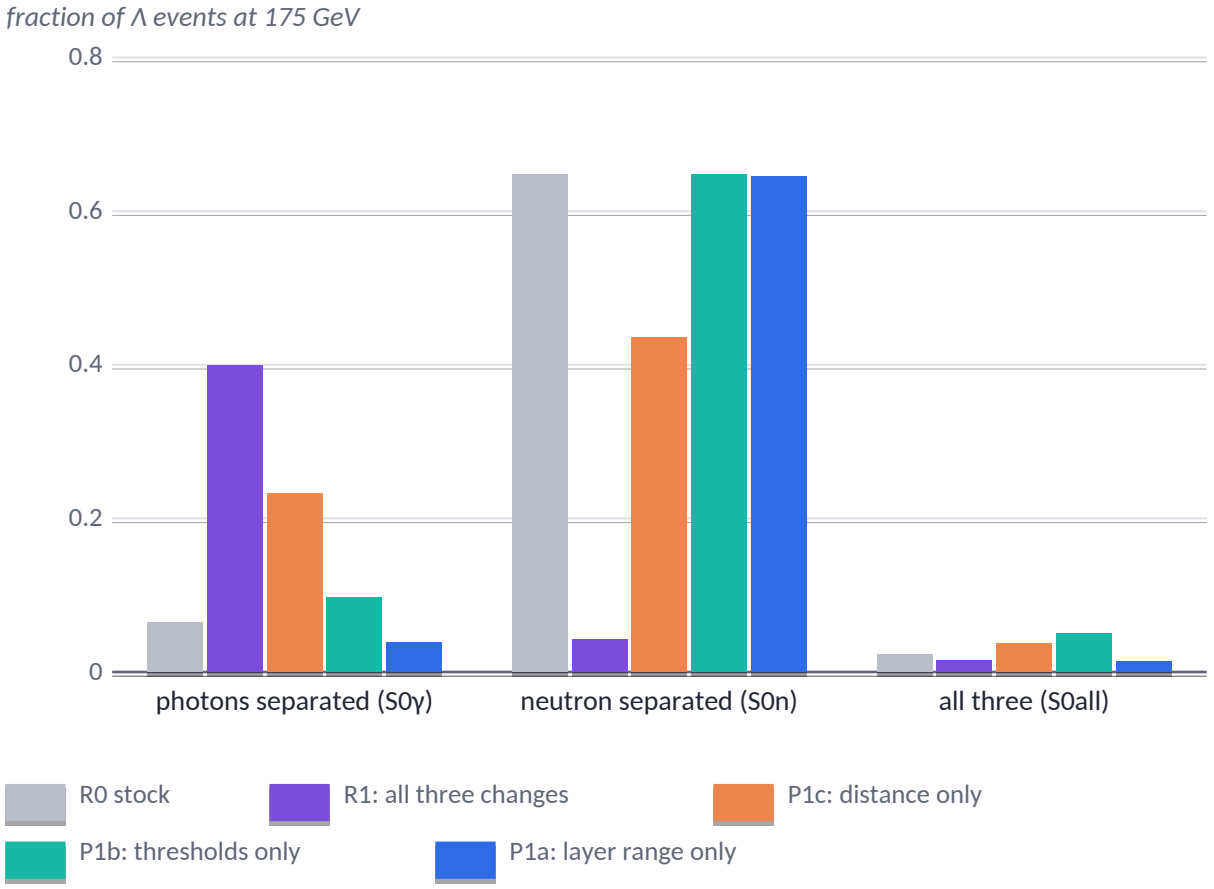


measured on the reconstruction — ceiling: computed offline with perfect separation and the measured resolution

- units: per cent of all thrown Λ → nπ⁰ at that momentum, so the geometric acceptance (1.6–33 %) is the ceiling of every bar
- left: the clustering moved separation from 0.3 to 0.6 of 17 points; the ceiling of this bar is not known (slide 8, first row)
- right: no 3×3 setting reconstructs a Λ; with the pairing as coded even perfect clusters give 0.9 %; a vertex-aware pairing gives 14 %
- at 275 GeV alone (the headline momentum): Λ ceiling 27.8 % of thrown Λ, acceptance 32.6 %; the current best reconstructs 0.00 %

Of the 17 points available, the clustering knobs moved 0.3; the pairing code holds 13.

What went well: PR 3064 split into its three changes, each measured alone



change in PR 3064	photons	neutron	verdict
neighbour distance 36.6 → 14.6 mm	+0.17	−0.21, clusters collapse to columns	rejected
seed 25 → 5 MeV, size 10 → 2 hits	+0.03	unchanged	accepted
layer range 1 → 3	−0.02	unchanged	rejected

- the PR's photon gain comes from the shorter distance; its neutron cost comes from the same distance
- the thresholds alone give a smaller gain and cost nothing: the keepable part of the PR
- the layer range makes it worse; one run closed that question

A reviewer of PR 3064 can now say which line to merge and which to drop, with numbers.

What went well: predictions before each run, paired tests, a held-out set that reproduced

Pre-registration and statistics

- every hypothesis timestamped in the ledger before its reconstruction started; verdicts of rejected runs kept
- noise floor measured first: 26 metric tests over 3 seeds; the quoted errors are honest, 3σ is the line
- paired test on identical events: gained − lost, so a +0.01 change is resolvable with 2000 events
- the ledger scores each prediction ✓ / ✗ against the result (slide 15)

E01: P1b against stock on the held-out sample (seed never used to choose)

momentum	search set Δ S0all	held-out Δ S0all	held-out gained / lost	z
125 GeV	+0.056	+0.058	116 / 0	10.8
175 GeV	+0.028	+0.031	62 / 0	7.9
250 GeV	+0.017	+0.012	23 / 0	4.8

- criterion written before the numbers existed: significant and positive at ≥ 2 of 3 momenta, nothing falls
- result: 3 of 3, zero events lost; the search-set gain repeats within errors

- **Also kept honest by construction**
  - a gate compares every non-ZDC collection to the reference file by file: a ZDC change that touched anything else would be rejected without appeal
  - rulers frozen per phase; the two ruler changes of the run (containment definition, raw-sum guard) are logged decisions with the user's words

The accepted gain is real: it repeated on events that played no part in choosing it.

What went well: nine defects found in code the program does not own

finding	what it is, in plain words	effect
F4	the Λ finder pairs photons as if they came from the collision point; only decays within ~3.6 m pass its π^0 mass window	caps the Λ yield at 0–7 % of accepted events, whatever the clustering does
F9	the decay-vertex iteration diverges for decays far from the collision point (hidden behind F4)	any fix of F4 exposes it; a stable vertex search measured offline reaches the acceptance
F6	the energy calibration does not fit the clusters it is applied to	photons read 13–31 % low
F7	a photon depth cut is measured from the ZDC centre, 1.3 m behind the front face the paper uses	removes real photons
F8	the photon shape cuts reject every ZDC photon on the new tiles	fix proposed in PR 3114 (reference R2 here)
F1, F2, F3, F5	stale collection names in the benchmarks; a unit fallback; subcell dimensions; no Λ on ZDC-only geometries	silent empty plots; workarounds in place

- all nine are open issues for EICrecon or the benchmarks, each with the file and line, a measurement and a proposed fix
- F4 came from the expert review of the loop's first version; the loop measured its size in Phase 0 on its own
- F9 was found only after the user asked for the acceptance (slide 14)

Two of the nine (F4, F9) bound the Λ yield more than any clustering setting in the loop's search space.

What went wrong: the loop tuned the clustering while the Λ ceiling sat in the pairing code

What the loop knew, and when

- Phase 0, night: measured the cap — with perfect inputs the pairing as coded passes 0–6.8 % of accepted Λ events (F4); filed as a finding, loop continued with its phase order
- Phases 1–2: 11 reconstructions on clustering knobs; separation ×3, Λ reconstructed 0 → 0
- 21:10 UTC, user asks about the acceptance; 21:55: an offline check with a vertex-aware pairing passes 73–85 % of accepted events (F9) — 10× the cap, independent of the clustering
- the loop's own backlog had that pairing as a Phase 3 experiment, not as a ceiling to measure first

0.9 %

Λ per thrown Λ → nπ^0 with perfect clusters and the pairing as coded (mean, 100–275 GeV)

14.1 %

same clusters, vertex-aware pairing (acceptance 16.9 %)

- **Why it happened**
  - the selector (S0) was chosen because the Λ count could not rank settings; that choice also disconnected the loop from the deliverable
  - the program's phase order (parameters, then code) was followed after the cap was known; no rule said 'measure every ceiling before tuning under it'
  - the expert review had estimated the cap at 10–30 %; the loop measured 0–7 % and did not re-plan on the difference

The cap was known on day one; the lever was measured only when the user pointed. Rule for run 2: ceilings first.

What went wrong: pre-registered predictions were right 69 % of the time

experiment	change	right / wrong	the miss
P1a	layer range 1 → 3	2 / 2	photon loss not predicted
P1b	seed 5 MeV, size 2 hits	2 / 3	photon gain 3–9× larger than predicted
P1c	distance 14.6 mm	3 / 3	half the photon gain; neutron did not collapse fully
P1d	R1 with layer range 1	1 / 3	cluster count went down, predicted up
P1e	R1 with stock thresholds	4 / 1	one magnitude miss
P1f	R1 with stock distance	5 / 0	—
E02	chaining 315 → 480 keV	5 / 4	neutron energy −13 % (predicted −5 %); low-momentum gain not predicted
E03	cluster floor 50 → 20 MeV	4 / 4	debris clusters on DIS events 5× the prediction
L01	P1b + PR 3114 photon cuts	7 / 0	—
E04	chaining 315 → 400 keV	7 / 0	— (the 'linear cost' branch, pre-registered, held)
E05	seed 5 → 2 MeV	5 / 0	— (the predicted null)
total		45 / 20	69 % of predicted cells right

- directions were mostly right; the misses are magnitudes, and most say the same thing: the cost was larger than the share measured on the clusters beforehand
- the hit rate rose through the run: Phase 1 17 / 12, Phase 2 28 / 8 — the later hypotheses were written with the earlier misses in hand
- a loop that mispredicts a third of its cells is screening the space, not yet modelling the detector; the tally belongs in every checkpoint report

Pre-registration made the miss rate measurable. Without it this slide could not exist.

What went wrong: the parameter knobs cannot separate what the showers merge

Reach — R1, P1c

- a shorter neighbour distance stops the chaining: photons separated 0.40 instead of 0.06
- the same distance cuts the neutron shower into single-column pieces: neutron separated 0.04 instead of 0.65
- cluster shapes on physics events fail the guards

Chaining threshold — E02, E04

- hits below 480 keV no longer join a cluster: the low-energy bridges between photon and neutron are cut
- S0all +0.015 at 175 GeV, significant at all 8 momenta
- the same hits are the neutron's own halo: its summed energy drops 13 % at 20 GeV (limit 5 %) → rejected
- E04, half the step (400 keV): half the gain, half the cost (−7.6 %) → rejected; family closed

Survival thresholds — E03, E05

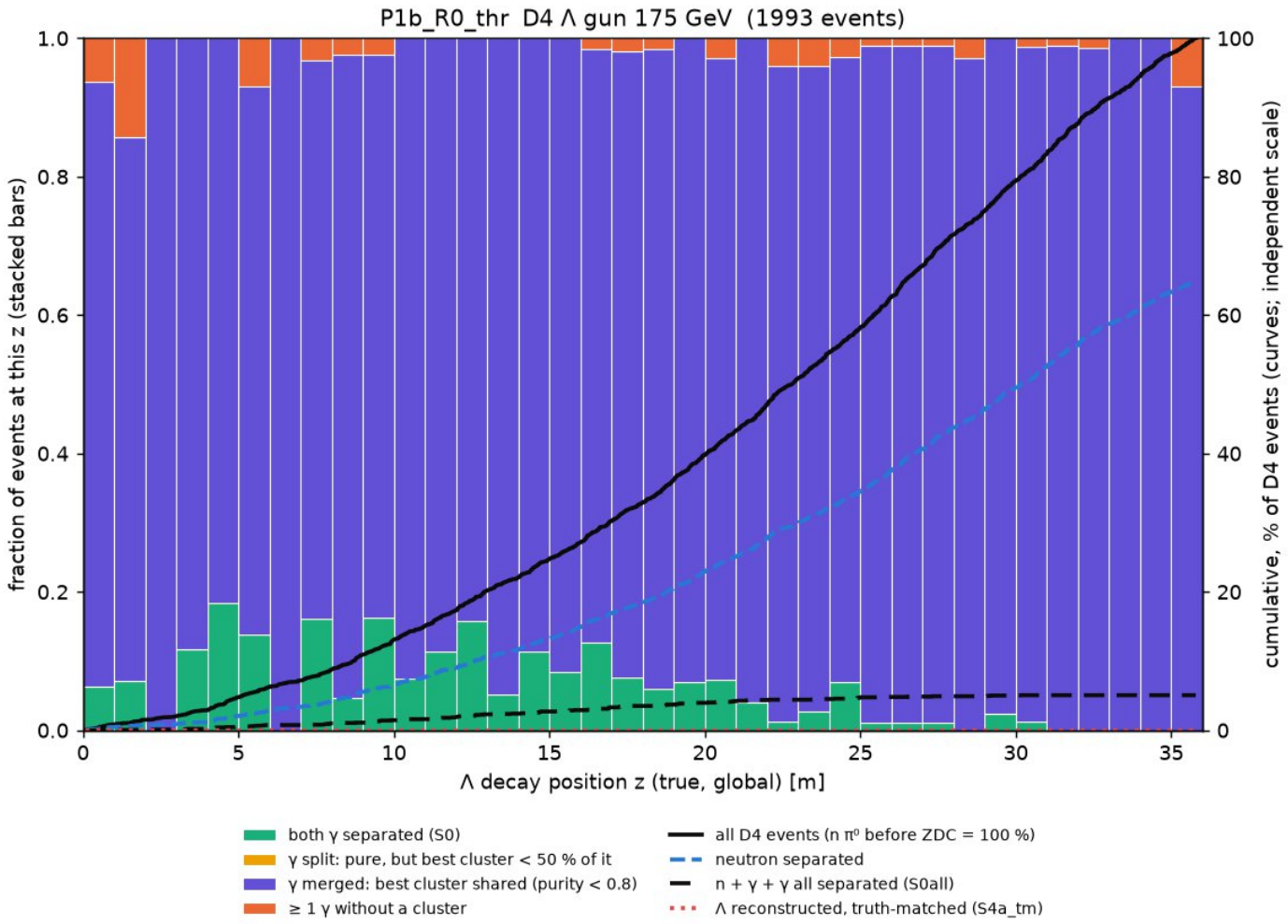
- clusters below 20 MeV deposited (≈ 0.7 GeV) now survive: the softest Λ photons get a cluster
- S0all +0.013 at 100 GeV; nothing above 150 GeV
- a 1 GeV photon and a 1 GeV neutron fragment look alike to a threshold: 0.97 debris clusters per DIS event (limit 0.5) → rejected
- E05, seed 5 → 2 MeV: no event changes → family closed

- **Where the remaining gain is**
  - a connected-component clusterer joins or cuts; it cannot split a dense photon core out of the neutron's halo
  - code changes, planned for Phase 3: split clusters at local maxima (the Island algorithm with splitting); chain hits at a high threshold, then collect the low ones onto the nearest cluster
  - the loop's parameter phase had a low ceiling by construction; ten of its twelve judged runs were threshold or distance steps, and two families are now closed by their guards

the code changes are planned, not run

After six Phase 2 runs both threshold families are closed by their guards; the next step is a code change.

Where the photons go: merged with the neutron shower at every decay position



- **How to read it (P1b, Λ at 175 GeV, 1993 events)**
  - bars, left axis: at each true decay position, the share of events where both photons are separated (green), share one cluster with something else (purple), or one photon has no cluster at all (orange)
  - curves, right axis, independent scale: cumulative share of events decaying before that z — all events reach 100 %; the neutron is separated in 65 %; all three in 5 %; Λ reconstructed: 0, on the axis
  - 87–97 % of photon pairs share a cluster at every z; early decays do slightly better
- **Why**
  - the π^0 carries ~15 % of the Λ energy; its photons land within the neutron's shower, and the clusterer joins any chain of neighbouring hits

The loss is one mechanism: photon showers chained to the neutron shower. Thresholds trim it; they cannot cut it.

What went wrong: operations — a full disk, silent truncation, a crash under load

time (UTC)	what happened	consequence
17:37	/data reached 100 % (865 GB of outputs, 63 GB per experiment)	two analyses, one reconstruction and one Λ count failed
18:50	failure noticed, by reading the logs 73 minutes later	no watchdog; empty log files looked like 'not started'
18:57	user decision: delete the rejected runs' output (315 GB); retention rule from now on	run resumes
19:05	8 output files cut short had been renamed as complete (the reconstruction exits 0 when writes fail)	a rerun would have skipped them and scored truncated data
11:53	one chunk crashed in a different detector's code while 24 jobs shared the machine	rerun alone, fine
earlier	ledger timestamps written ahead of the clock	corrected by hand

Root cause

- every experiment writes every collection — full tracking on deep-inelastic events — for a gate that a ZDC parameter cannot fail
- no preflight check of free space; no integrity check of outputs

Fixed during the run / still open

- an output now needs its per-file record or it is redone; rejected runs deleted after the verdict; ≥ 150 GB before a launch
- open: restrict the collections written (63 GB → ~5 GB per experiment)

Unattended loops fail on plumbing; run 1's plumbing had no checks at all.

Changes for run 2, ranked by leverage

#	change	expected effect
1	a selector tied to the Λ chain: run the vertex-aware pairing offline on each setting's real clusters	a number that moves with the Λ yield, minutes per experiment, no EICrecon change
2	an ideal-partition ceiling for separation: give each subcell to the particle that put most energy in it, score with the same rule	a denominator for every separation table
3	write only the ZDC collections for parameter runs; run the full gate once per code build	63 GB → ~5 GB, 2.5 h → ~40 min per experiment; scans instead of single steps
4	code tier as soon as a parameter family is closed by its guards	the remaining separation gain (split at local maxima; chain high, collect low)
5	preflight and integrity checks in the launcher: free space, output records, non-empty logs, error scan	failures found in minutes, by the loop
6	a prediction tally and the cost per experiment in every checkpoint report	the loop reports its own hit rate and its budget
7	loop rule: measure every ceiling before tuning under it; re-plan when a measured ceiling contradicts the plan	no proxy optimisation while the cap is elsewhere

all seven are proposals, none implemented yet

Items 1, 2 and 7 would have re-aimed run 1 by its second hour.

Status at the stop: one accepted, three rejected, two inconclusive, two families closed

experiment	change	verdict	reason
E01 = P1b	seed 5 MeV, size 2 hits	ACCEPTED	held-out check reproduced; current best
E02	chaining threshold 480 keV	REJECTED	neutron energy −13 % at 20 GeV (limit 5 %)
E03	cluster floor 20 MeV	REJECTED	0.97 debris clusters per DIS event (limit 0.5)
L01	P1b + PR 3114 photon cuts	INCONCLUSIVE	25 Λ gained, 0 lost in 16 000 events; no momentum at 3σ
E04	chaining threshold 400 keV	REJECTED	neutron energy −7.6 % at 20 GeV (limit 5 %); chaining family closed
E05	seed 2 MeV	INCONCLUSIVE	no event changes, as predicted; survival-threshold family closed

Decisions pending, for the user

- E04 / E05 output (126 GB) falls under the retention rule once the user confirms
- adopt the run-2 changes, in particular the Λ-chain selector before any further clustering run
- Phase 3 code tier next (split at local maxima; chain high, collect low), or a re-planned Phase 2

What exists, reproducibly

- every number on these slides rebuilds from space/perf/*.json and the ledger (slides/build_autoresearch_retro_v1_slides.py)
- every experiment is regenerable from its config at the pinned EICrecon build
- transcripts, decisions and questions are kept for the write-up

Run 1 produced a clear negative result on the clustering knobs, nine findings, and the measured location of the real lever.